Etiology and Pathogenesis

Etiology of capillary malformations (CM) is currently unknown. Facial CM is thought to result from clonal expansion of abnormal cells originating from the neural crest. The identification of the affected gene (*RASA1*) in CM-AVM (see Arteriovenous Malformations) may provide insights into the pathogenesis of the more common sporadic CM.

Histologically, CM is characterized by dilatation of a normal number of capillaries in the papillary and upper reticular dermis, along with areas showing an increased number of otherwise normal-appearing capillaries (see Fig. 173-2). Endothelial cells are flat. Factor VIII, fibronectin, and basement membrane proteins are normal, but S100 staining reveals abnormal innervation.

CAPILLARY MALFORMATIONS AT A GLANCE

- **Worldwide occurrence**: Affects approximately 0.1% to 0.2% of the population
- **Type**: Congenital no-flow malformations of the capillary bed
- **Color**: Pinkish-red to purple; tends to darken and thicken over time
- **Clinical significance**: In most cases, a cosmetic concern
- **Syndromic associations**: Can be part of Sturge-Weber syndrome, Klippel-Trénaunay syndrome, or phakomatosis pigmentovascularis
- **Pathology**: Capillaries are increased in size and number, with evidence of abnormal innervation

Clinical Findings

CUTANEOUS LESIONS (Fig. 173-3; See Fig. 173-2)

Capillary malformations (CM) present as red, homogenous, congenital lesions that are typically unilateral, occasionally bilateral, and rarely midline. They involve the skin and subcutis, and sometimes mucosa.

- **Color**: Ranges from pinkish-red to deep purple
- **Shape**: Geographic borders or dermatomal distribution
- **Texture**: Flat, painless, non-blanching (under pressure), do not bleed spontaneously, and are not warm to touch
- **Distribution**: Can occur anywhere on the body; 50% are located on the face, following the trigeminal nerve branches:
- **V1 (ophthalmic)**: Forehead, upper eyelid
- **V2 (maxillary)**: Lower eyelid, cheek, upper lip
- **V3 (mandibular)**: Lower lip, chin, mandible

CM may be associated with other vascular malformations as part of a syndrome, such as: - **Sturge-Weber syndrome** - **Klippel-Trénaunay syndrome** - **Servelle-Martorell syndrome** - **Phakomatosis pigmentovascularis**

These syndromes are not inherited.

Rarely, CM in the lumbosacral region may be a cutaneous marker of occult spinal dysraphism.

Sturge-Weber Syndrome

(Section continues...)